



Nida Fathima

Calicut, India

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Technical Skills

Molecular Biology Techniques:

PCR

Transcription

Gel Electrophoresis (Agarose, Native PAGE, Denaturing PAGE, SDS PAGE)

DNA/RNA purification

Nucleic Acid Analysis:

Aptamer design using NUPACK

Fluorescence analysis:

Synergy H1 microplate reader

UV-Melting techniques

Spectroscopy & Particle Analysis:

Circular Dichroism (CD) Spectroscopy

Dynamic Light Scattering (DLS)

Computational & Data Analysis:

Python programming

Data handling in Excel

Presentation & Communication:

Strong scientific reporting and presentation skills

Achievements

GATE Qualified (2025)

Volunteering

National Service Scheme (2018 - 2023)

Bharat Scout & Guide (2010 - 2017)

Languages

English - Fluent

Malayalam-Fluent

Hindi - Intermediate

Arabic - Intermediate

resume(pdf)

Profile

Passionate researcher with experience at the intersection of chemistry and biology, with a growing interest in data analysis. Experienced in designing and characterizing RNA/DNA nanostructures for drug delivery and nanotechnology applications. Strong analytical, computational, and presentation skills, with expertise in Python programming and data handling.

Research Experience

MSc Thesis

"Engineering of Highly Stable RNA Nanomotifs for Nanotechnology Applications"

- Supervisor : Dr. Ashwani Sharma

Indian Institute of Science Education and Research, Tirupati

June 2022 - May 2023

- Designed a fluorescence-based method to compare the stability of different RNA three-way junctions (3WJs) used in RNA nanotechnology.

- Investigated the potential of RNA nanostructures in targeted drug delivery applications.

Semester Project

"Construction of Stimuli-Specific DNA Nanostructures Using i-Motif Sequences"

- Supervisor : Dr. Ashwani Sharma

Indian Institute of Science Education and Research, Tirupati

Jan 2022 - Apr 2022

- Designed pH-responsive DNA nanostructures such as triangles, squares, and tetrahedrons using i-motif sequences.

Education

BS-MS Chemistry

Indian Institute of Science Education and Research, Tirupati

2018 - 2023

7.0 CGPA

Contact Reference

- Dr. Ashwani Sharma, Assistant Professor, Department of Chemistry and Biology, IISER Tirupati, ✉ a.sharma@labs.iisertirupati.ac.in
- Dr. Pankaj Kumar Koli, Assistant Professor, Department of Chemistry, IISER Tirupati, ✉ pankaj@labs.iisertirupati.ac.in

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